

btctax interview walk — TY2024 / TY2025

Lived journey walk — the btctax interview, played as a real filer

Two walks, both driven through the real CLI (`init` → `import` → `income` → `import/income` → `answer` → `report` → `export-irs-pdf`), never the TUI. Every command below actually ran against the binary built from this tree (`cargo build`, dev profile, at commit `121c8805`). All figures are invented; every SSN is in the never-issued area-000 space; every EIN/TIN is drawn from `scripts/pii-scan-generic.sh`'s `ALLOWED_EIN` list; every name is obviously fictional (surname “Testcase”).

Working directories (scratch, nothing committed): `/tmp/claude-1000/-scratch-code-bitcoin-tax/d76d9057-52e8-4b29-bf10-a27f4b73464c/scratchpad/journey/{walkA,walkB,probe-carloan}`.

1. The household

Robin & Dana Testcase, Married Filing Jointly, TY2024. Robin works; Dana does not (no income).

Item	Value
Taxpayer	Robin Testcase, SSN 000-11-2222, DOB 1985-04-12, Software Engineer
Spouse	Dana Testcase, SSN 000-11-3333, DOB 1987-09-30, Homemaker (no income)
Address	742 Ledger Lane, Springfield, IL 62704
W-2 wages (Robin, Ledger Analytics Inc., EIN 56-1234567)	Box1 \$800,000 · Box2 fed w/h \$200,000 · Box3 SS wages \$168,600 · Box4 SS w/h \$10,453.20 · Box5 Medicare wages \$800,000 · Box6 Medicare w/h \$17,000 · Box12 code W (HSA, employer+payroll) \$3,000 · Box17 state w/h \$40,000
Interest (1099-INT, Frontier Savings Bank, TIN 11-1111111)	Box1 \$2,500
Dividends	none

Item	Value
Dependents (4, all qualifying children living with both parents all year, all U.S. citizens)	
— Nova Testcase, daughter	SSN 000-11-4444, DOB 2020-05-14 (age 4 — under 6)
— Milo Testcase, son	SSN 000-11-5555, DOB 2016-09-02 (age 8)
— Ivy Testcase, daughter	SSN 000-11-6666, DOB 2012-11-20 (age 12)
— Jasper Testcase, son	SSN 000-11-7777, DOB 2006-08-05 (age 18 — 17+, over the CTC age cutoff)
HSA (family HDHP coverage, both spouses covered under Robin’s plan)	Employer/payroll (W-2 box12 W) \$3,000 + Robin’s own out-of-pocket (Form 8889 line 2) \$2,000 = \$5,000 total, under the \$8,300 TY2024 family limit
Mortgage (Form 1098, Prairie Home Mortgage Co., TIN 12-3456789)	Box1 interest \$40,000 · Box2 outstanding principal \$680,000 · originated 2021-03-01 (under the \$750,000 §163(h)(3)(B) post-2017 acquisition-debt limit)
Student loan (Form 1098-E, Federal Direct Loan Servicing, TIN 98-7654321)	Box1 interest \$2,700 (paid; §221 phases out to \$0 at this income — expected)
SALT	Real estate tax \$12,000 + state income tax withheld \$40,000 (both real, capped at \$10,000 by §164(b)(6))
Car loan	\$1,850 interest paid in 2024 on a 2025 truck loan — see Finding #1 : TY2024 has no field for it at all
Bitcoin (Coinbase, all long-term)	see table below

Bitcoin lots and disposals (walkA/coinbase.csv, a synthetic Coinbase “Transactions” export)

Date	Type	BTC	Price	Amount
2021-06-01	Buy	1.20000000	\$36,584.53	\$43,901.44
2022-11-01	Buy	0.80000000	\$20,517.05	\$16,413.64
2023-08-01	Buy	0.50000000	\$29,227.14	\$14,613.57
2024-09-01	Sell	0.40000000	\$58,595.27	proceeds \$23,408.11 (after \$30 fee)
2024-10-15	Sell	0.30000000	\$66,236.66	proceeds \$19,846.00 (after \$25 fee)
2024-11-01	Sell	0.50000000	\$69,715.15	proceeds \$34,817.58 (after \$40 fee)

Date	Type	BTC	Price	Amount
2024-12-05	Sell	0.30000000	\$100,954.70	proceeds \$30,251.41 (after \$35 fee)

All prices are the bundled dataset’s real daily close (crates/btctax-adapters/data/btc_usd_daily_close.csv). Every buy predates the earliest sale by ≥ 13 months, so every disposal is long-term regardless of which lot FIFO consumes — verified: report --tax-year 2024 (crypto-only, before any full-return data) showed Part I (short-term): 0.00 / Part II (long-term): proceeds 108323.10 cost basis 52669.58 gain 55653.52.

2. Walk A — TY2024, step by step

2.1 Build

```
$ cargo build --workspace
  Compiling btctax-core v0.18.0 ...
  Compiling btctax-cli v0.18.0 ...
  Compiling btctax-tui-edit v0.18.0 ...
  Finished `dev` profile [optimized + debuginfo] target(s) in 14.94s
```

2.2 btctax init

init reads the vault passphrase from a secure prompt (no TTY → refuses immediately with error: io: No such device or address (os error 6) — this is a **deliberate** secret-handling refusal, not a bug; per this project’s severity ruling secret-handling friction never gates). Driven interactively via a pty (pexpect):

```
$ btctax --vault v.pgp init --key-backup key-backup.asc
Vault passphrase: *****
Confirm passphrase: *****
Initialized vault v.pgp (key backed up to key-backup.asc)
```

For every command after this, the documented BTCTAX_PASSPHRASE env var (main.rs:51) was exported so later commands run non-interactively — the same mechanism the tool’s own docs recommend for scripting.

2.3 Import the Bitcoin ledger

```
$ btctax --vault v.pgp import coinbase.csv
Import:
  coinbase [coinbase.csv]: parsed 7 rows → 7 BTC events (0 dropped
no-BTC, 0 unclassified)
  appended 7 | duplicates 0 | NEW import-conflicts 0
```

```
$ btctax --vault v.pgp verify
Conservation (FR9): BALANCED
  in 250000000 = disposed 150000000 + removed 0 + held 100000000 +
fee-sats 0 + pending 0
2025 transition: Path A (actual per-wallet reconstruction; default, no
election)
Pending reconciliation: 0 transfer(s); unknown-basis inbounds: 0
Hard blockers (gate tax computation): 0
Advisory blockers: 1
[Pre2025MethodNote] import|coinbase|trade|cb-sell1 :: pre-2025 lots
reconstructed under FIFO (FIFO is the §7.4 legal default); you have
NOT declared your filed pre-2025 lot method — if your filed pre-2025
returns used a different method your carryforward basis may differ.
Declare it: config --set-pre2025-method <m> --attest-pre2025-method
Pre-2025 method (attested historical fact): FIFO (attested: false)
```

Followed the tool's own instruction:

```
$ btctax --vault v.pgp config --set-pre2025-method hifo --attest-
pre2025-method
fee_treatment: non-taxable, basis carries (TP8 c)
pre2025_method: FIFO (attested: true)
forward_method: FIFO (vault-wide, in force as of 2026-09-09)
```

Crypto-only report before any household data is imported, confirming the disposals land as I designed them:

```
$ btctax --vault v.pgp report --tax-year 2024
TY2024 — filable (20 forms; TaxTable yes; full-return params yes;
1099-DA none)
Federal tax attributable to crypto — tax year 2024
  NOT COMPUTABLE [TaxProfileMissing]: no tax_profile set for 2024
Schedule D (raw pre-netting part totals) — tax year 2024
  Part I (short-term): proceeds 0.00   cost basis 0.00   gain 0.00
  Part II (long-term):  proceeds 108323.10   cost basis 52669.58
gain 55653.52
```

2.4 The household TOML (income import)

Per the spec's own §4.2 decision, box-level document data (W-2/1099/1098 amounts) and dependent identity/§152-gate leaves have no CLI form-entry screen — they enter through income import's TOML or the TUI, exactly as the tool's own worked example (docs/examples/examples.md J6) does it. I built household.toml with every dollar figure, every identity field, and the full dependent §152 gate set (matching the shape of the shipped fixture crates/btctax-cli/tests/fixtures/examples/nine_dependents_amt_inputs.toml), but **deliberately omitted** every “fail-loud, no safe default” gate — the whole [documents] census, home_sale.sold_main_home, can_be_claimed_as_dependent_{taxpayer,spouse}, foreign_accounts,

foreign_trust, sch1.hsa_activity, dual_status_alien, filing_form_4952, other_out_of_scope_income, digital_asset_activity, claiming_mortgage_interest_credit, filer_tin_issued_by_due_date, the three mortgage declarations, all nine HSA (Form 8889) eligibility gates, and the skippables (blindness ×2, DOBs, the sales-tax election, donation questions) — so that those would be asked LIVE by income answer, exercising the real interview.

```
$ btctax --vault v.pgp income import --year 2024 --file household.toml
Imported full-return inputs for tax year 2024.
```

2.5 income answer — the live interview

```
$ btctax --vault v.pgp income answer --year 2024
Answering full-return questions for tax year 2024:
interview: 36 question(s) still to answer · return: computable
```

— Step 0: your ledger, for 2024 —

LEDGER BLOCKERS (1):

- 1 × Pre2025MethodNote (advisory — it never gates, and it is still worth clearing)

→ btctax config --set-pre2025-method <m> --attest

IMPORTS BY VENUE (1):

- coinbase: 7 imported event(s)

VENUES vs FORM 1099-DA ANSWERS (1):

- TY2024's Form 1099-DA regime reports nothing (no Form 1099-DA is issued for this year) — no Form

1099-DA question is asked for this year and an answer would be refused as unread, so your

disposals on coinbase are boxed by mechanism, not by an answer

STANDING ORDERS (§1.1012-1(j) — outside Notice 2026-20's relief period) (1):

- exchange:coinbase:default: your first custodial disposition of 2024 is dated 2024-09-01, and no

dated method election was in force for it. Notice 2026-20's relief period BEGINS 2025-01-01

(§3.03), so §4.02(2)'s standing-order relief does not reach this sale, and btctax refuses an

election effective before that date as back-dated — the pre-2025 method is a declaration you

attest to instead

→ btctax config --set-pre2025-method <m> --attest

— The answer panel (before) —

BLOCKING (36) — commit waits on these: [36 bulleted prompts, one per live class-(A) question —

every document census row, the HSA/mortgage/foreign/digital-asset/dual-status declarations,

filer-TIN, spouse-election, mortgage-credit and sold-main-home]
 FORGOING (9) – lawful to skip: blindness ×2, the sales-tax election,
 both DOBs, both death dates,
 the donation-restriction and CWA questions
 NOT COMPUTED (1): child tax credit / credit for other dependents – 4
 dependents, up to \$2000 each
 (59 answered, 66 not applicable to this return)

Then it asked, one line at a time. The **document census** (verbatim, in the order asked — this is the FIRST round, and note it does **not** follow the panel's print order):

Did you receive one or more Form W-2 (every employer must send you one – Form 1040 line 1a instructions)? [y/n]: y
 Did you receive one or more Form 1099-INT (each payer should send you one – Form 1040 line 2b instructions)? [y/n]: y
 Did you receive one or more Form 1099-DIV (each payer should send you one – Form 1040 line 3b instructions)? [y/n]: n
 Did you receive one or more Form 1099-B (each broker should send you one – Form 8949 and Schedule D instructions)? [y/n]: n
 Did you receive one or more Form 1099-G (a state that refunded income tax, or paid unemployment compensation, should send you one – Schedule 1 line 1 and line 7 instructions)? [y/n]: n
 Did you receive one or more Form 1098 (your mortgage lender should send you one – Schedule A line 8a instructions)? [y/n]: y
 Did you receive one or more Form 1098-E (your student loan servicer should send you one – Schedule 1 line 21 instructions)? [y/n]: y
 Did you receive one or more Form 1099-R (the payer of an IRA, pension or annuity distribution must send you one – Form 1040 lines 4a-5b instructions)? [y/n]: n
 Did you receive a Form SSA-1099 or RRB-1099 ...? [y/n]: n
 Did you receive one or more Form 1099-NEC, Form 1099-MISC or Form 1099-K ...? [y/n]: n
 Did you receive one or more Schedule K-1 ...? [y/n]: n
 Did you receive rent or royalties from real estate or from property you own ...? [y/n]: n
 Did you receive a Form 1099-S ...? [y/n]: n
 Did you receive one or more Form 1099-OID ...? [y/n]: n
 Did you receive one or more Form W-2G ...? [y/n]: n
 Did you receive one or more Form 1099-C ...? [y/n]: n
 Did you receive a Form 1095-A ...? [y/n]: n
 Did you receive a Form 1098-T ...? [y/n]: n
 Did you receive one or more Form 1099-SA (the trustee of a health savings account must send you one for any distribution – Form 8889 line 14a instructions)? [y/n]: n
 Did you receive one or more Form 5498-SA (the trustee of a health savings account sends one reporting the year's contributions and the account's fair market value ...)? [y/n]: n

Then the return-level declarations (same first round, interleaved by registry order — a representative slice):

Can someone claim YOU as a dependent on their return? [y/n]: n
 Can someone claim YOUR SPOUSE as a dependent on their return? [y/n]: n
 Schedule B line 7a: did you have a financial interest in, or signature authority over, a FOREIGN financial account? [y/n]: n
 Schedule B line 8: did you receive a distribution from — or were you the grantor of, or transferor to — a FOREIGN TRUST? [y/n]: n
 In this tax year, did ANY of these happen with a health savings account? — (a) anyone ... put money into one for you; (b) you took money out of one; (c) you inherited one; or (d) you stopped being HSA-eligible ... [y/n]: y
 Were you a DUAL-STATUS ALIEN this year ...? [y/n]: n
 Did you use ALL of your home-mortgage loan(s) to buy, build, or improve that home? (Schedule A line 8: if not, the box is checked.) [y/n]: y
 Is the home your Form 1098 mortgage interest relates to a principal residence ...? (Form 6251 line 3 — a houseboat or recreational vehicle is NOT an AMT-qualified dwelling.) [y/n]: y
 In this tax year, did ANY of these happen? (a) You received income other than what you have entered here ... [huge catch-all, verbatim ~2,900 words] [y/n]: n
 Schedule A line 8a — were you inside EVERY home-mortgage debt limit this year? ... [y/n]: y
 Are you filing Form 4952 (Investment Interest Expense Deduction)? ... [y/n]: n
 At any time during 2024, did you: (a) receive (as a reward, award, or payment for property or services); or (b) sell, exchange, or otherwise dispose of a digital asset (or a financial interest in a digital asset)? (Form 1040 instructions, Digital Assets: holding a digital asset; moving one between wallets or accounts you own or control; and purchasing digital assets with U.S. or other real currency do NOT, alone, require a "Yes".) [y/n]: y
 Did you, and your spouse if filing a joint return, have either an SSN or ITIN issued on or before the due date of your return (including extensions)? ... [y/n]: y
 Was your spouse a nonresident alien or a dual-status alien at any time during the tax year? ... [y/n]: n
 Schedule A line 8a — are you claiming the MORTGAGE INTEREST CREDIT? (Form 8396 ...) [y/n]: n
 Did you SELL OR EXCHANGE YOUR MAIN HOME during the year? (Schedule D, "Sale of Your Home" ...) [y/n]: n

Dependent gates — asked for **every** dependent, **every leaf**, even though every one of these 4×~17 leaves was already truthfully answered in the imported TOML (see Finding #3). One full child shown verbatim, the other three identical in shape:

[Nova Testcase] What is this person's date of birth? [YYYY-MM-DD; currently 2020-05-14]: <Enter>

[Nova Testcase] Is this person your son, daughter, stepchild, foster child, brother, sister, stepbrother, stepsister, half brother, half sister, or a descendant of any of them (for example, your grandchild, niece, or nephew)? [y/n, currently y]: <Enter>

[Nova Testcase] Was this person younger than you (or your spouse if filing jointly) at the end of the tax year? [y/n, currently y]: <Enter>

[Nova Testcase] Was this person a full-time student? (...) [y/n, currently n]: <Enter>

[Nova Testcase] Was this person permanently and totally disabled? (...) [y/n, currently n]: <Enter>

[Nova Testcase] Did this person provide over half of their own support for the tax year? (See Pub. 501.) [y/n, currently n]: <Enter>

[Nova Testcase] Is this person filing a joint return for the tax year? [y/n, currently n]: <Enter>

[Nova Testcase] Did this person live with you for more than half of the tax year? (Read the Exception to time lived with you before answering ...) [y/n, currently y]: <Enter>

[Nova Testcase] And did this person live with you in the United States? [y/n, currently y]: <Enter>

[Nova Testcase] Does this child meet the conditions to be a qualifying child of any other person (other than your spouse if filing jointly) for the tax year? [y/n, currently n]: <Enter>

[Nova Testcase] Was this person a U.S. citizen, U.S. national, U.S. resident alien, or a resident of Canada or Mexico? (...) [y/n, currently y]: <Enter>

[Nova Testcase] Was this person married? [y/n, currently n]: <Enter>

[Nova Testcase] Did this person have an SSN, ITIN, or adoption taxpayer identification number (ATIN) issued on or before the due date of your return (including extensions)? (...) [y/n, currently y]: <Enter>

[Nova Testcase] Was this person a U.S. citizen, U.S. national, or U.S. resident alien? (...) [y/n, currently y]: <Enter>

[Nova Testcase] Did you, or your spouse if filing a joint return, and this child have SSNs valid for employment and issued before the due date of your return (including extensions)? (...) [y/n, currently y]: <Enter>

(repeated identically, in order, for Milo, Ivy, then Jasper)

Then the skippables (lawful to skip; answered truthfully rather than skipped):

Are YOU legally blind? (§63(f) additional deduction) [y/n; Enter to skip]: n

Is YOUR SPOUSE legally blind? (§63(f) additional deduction) [y/n;

Enter to skip]: n
 Deduct general SALES taxes instead of state/local income taxes?
 (§164(b)(5)) [y/n; Enter to skip]: n
 YOUR date of birth [none; Enter to skip]: 1985-04-12
 YOUR SPOUSE's date of birth [none; Enter to skip]: 1987-09-30
 Did YOU (the taxpayer named on this return) die during the tax year?
 (...) [y/n; Enter to skip]: n
 Did YOUR SPOUSE die during the tax year? (...) [y/n; Enter to skip]: n
 Did ANY property you donated this year have strings attached? (...)
 [y/n; Enter to skip]: n
 For EVERY charitable gift of \$250 or more that this return DEDUCTS ...
 do you already hold ... a CONTEMPORANEOUS WRITTEN ACKNOWLEDGMENT ...?
 [y/n; Enter to skip]: y

And then — **not previewed anywhere in the “before” panel** (see Finding #4) — two questions became live mid-session, triggered by my own census answers a moment earlier:

Did you receive taxable interest or dividends for which no Form 1099-INT or Form 1099-DIV was issued — a bank paying under \$10, a seller-financed mortgage you hold, or a nominee distribution? (Schedule B line 1 ...) [y/n]: n
 Did you receive a refund, credit or offset of state or local income taxes in 2024? (Form 1040 instructions, Schedule 1 line 1 ...) [y/n]: n

And finally, because I had just answered “yes” to the HSA-activity question, the whole **Form 8889** gate set opened live, in the same session, no restart needed:

Form 8889 line 14a: did you take any money out of an HSA this year that no Form 1099-SA reports? (...) [y/n]: n
 Form 8889 line 1: was YOUR OWN high-deductible health plan (HDHP) coverage FAMILY coverage? (...) [y/n]: y
 Form 8889 line 3: on the first day of EVERY month this year, were you (or were you considered) an eligible individual with the SAME coverage? (...) [y/n]: y
 Form 8889 lines 3 and 7: were you age 55 or older at the END of this tax year? (...) [y/n]: n
 Form 8889 line 3: were you enrolled in MEDICARE for any month this year? (...) [y/n]: n
 Form 8889 Parts I, II and III: do BOTH you and your spouse each have a separate HSA? (...) [y/n]: n
 Form 8889 line 4: did you or your employer contribute to an ARCHER MSA this year, or did you have a Medicare Advantage MSA? (...) [y/n]: n
 Form 8889 Part III: did you STOP being an eligible individual during a testing period this year ...? [y/n]: n
 Form 8889 lines 1 and 3: did your SPOUSE have an HDHP with FAMILY coverage? (...) [y/n]: y

— The answer panel (after) —

NOT COMPUTED (1) – btctax does not file the schedule these are figured on; the credit boxes on your

return are printed, the amount is yours to enter:

- child tax credit / credit for other dependents not computed – 4 dependents with a credit box;

up to 2000 each

(115 answered, 55 not applicable to this return)

Answered the full-return questions for tax year 2024.

115 questions answered in one sitting, everything blocking cleared, the return committed (this command's own final write, not a draft — see §2.6).

2.6 The finished return — report --tax-year 2024

```
$ btctax --vault v.pgp report --tax-year 2024
```

```
TY2024 – filable (20 forms; TaxTable yes; full-return params yes;
1099-DA none)
```

```
Federal tax attributable to crypto – tax year 2024
```

```
net short-term: 0.00 net long-term: 55653.52
```

```
crypto ordinary income (level): 0.00
```

```
ordinary-rate tax (attributable): 0.00
```

```
LTCG tax (attributable): 11130.70 NIIT (attributable): 2114.83
```

```
TOTAL federal tax attributable to crypto (delta): 13245.53
```

```
marginal rates: ordinary 0.37 / LTCG 0.238 all-in
```

== Absolute filed return (Form 1040) – tax year 2024 ==

```
Total income (1040 L9): 858154.00
```

```
Adjustments (L10): 2000.00
```

```
AGI (L11): 856154.00
```

```
Deduction (L12, itemized): 50000.00
```

```
Taxable income (L15): 806154.00
```

```
Tax (L16): 214941.00
```

```
Additional Medicare (Form 8959 → Sch 2 L11): 4950.00
```

```
Net Investment Income Tax (Form 8960 → Sch 2 L12): 2210.00
```

```
Alternative Minimum Tax (Form 6251 → Sch 2 L2): 0.00
```

```
AMTI (L4) 816153.52 · exemption (L5) 133300.00 · tentative minimum
tax (L9) 182094.70 · regular tax (L10) 214941.00
```

Line 7 does not exceed line 10 → no Form 6251 attachment is required.

```
TOTAL TAX (L24): 222101.00
```

```
Total payments (L33): 205400.00
```

```
→ AMOUNT OWED (L37): 16701.00
```

— ADVISORIES (6) —

- CTC/ODC NOT COMPUTED, AND NOT AVAILABLE TO YOU – you captured 4

dependent(s), and v1 does not

compute the Child Tax Credit or the Credit for Other Dependents.

Here that costs you NOTHING:

your income phases the credit out entirely under §24(b) ... 1040
line 19 is \$0 and that is the

correct figure – there is no Schedule 8812 for you to file.

- OTHER CREDITS NOT COMPUTED ...
- DEDUCTIONS NOT COMPUTED ...
- RETURN OPTIONS NOT OFFERED ... (fiscal year, third-party designee,
apply-overpayment, refund split,
email, and the §6013(g)/(h) nonresident-spouse election)
- PRIOR-YEAR CARRYOVERS NOT STATED ...
- FORM 8960 LINE 9B NOT CLAIMED – ... your tax is currently
OVERSTATED by up to \$380 ...

(Advisories never change a number and never fail the command.)

== The interview – tax year 2024 ==

Documents received (the census)

Form W-2	YES	1 row(s)
transcribed		
Form 1099-INT	YES	1 row(s)
transcribed		
Form 1098	YES	1 row(s)
transcribed		
Form 1098-E	YES	1 row(s)
transcribed		

[... 15 more rows, all "no", 11 of them "no section holds rows of
this type" ...]

Sale of your main home

sold your main home during the year: no

DECISION: no sale to report

Dependents – row (7), the credit box the Form 1040 flowchart
computes

(1) Nova Testcase	Child tax credit
(2) Milo Testcase	Child tax credit
(3) Ivy Testcase	Child tax credit
(4) Jasper Testcase	Credit for other dependents

Where each figure came from (14 figure(s) with an amount)

form_1098[0].box1_interest	\$40000
Form 1098 #1 Prairie Home Mortgage Co. • TIN **-***6789	
form_1098[0].box2_outstanding_principal	\$680000
Form 1098 #1 Prairie Home Mortgage Co. • TIN **-***6789	
form_1098e[0].box1_interest	\$2700

```

Form 1098-E #1 Federal Direct Loan Servicing · TIN **-***4321
    hsa.line2_contributions_you_made                $2000
your own records (no information return reports it)
    int_1099[0].box1_interest                      $2500
Form 1099-INT #1 Frontier Savings Bank · TIN **-***1111
    schedule_a.salt_real_estate                    $12000
your own records (no information return reports it)
    w2s[0].box12[0].amount                         $3000
Form W-2 #1 Ledger Analytics Inc. · TIN **-***4567
    w2s[0].box17_state_tax_withheld                 $40000
Form W-2 #1 Ledger Analytics Inc. · TIN **-***4567
    w2s[0].box1_wages                              $800000
Form W-2 #1 Ledger Analytics Inc. · TIN **-***4567
    w2s[0].box2_fed_withheld                       $200000
Form W-2 #1 Ledger Analytics Inc. · TIN **-***4567
    w2s[0].box3_ss_wages                           $168600
Form W-2 #1 Ledger Analytics Inc. · TIN **-***4567
    w2s[0].box4_ss_withheld                        $10453.20
Form W-2 #1 Ledger Analytics Inc. · TIN **-***4567
    w2s[0].box5_medicare_wages                     $800000
Form W-2 #1 Ledger Analytics Inc. · TIN **-***4567
    w2s[0].box6_medicare_withheld                  $17000
Form W-2 #1 Ledger Analytics Inc. · TIN **-***4567

```

Transcribed without a date

Form 1099-INT #1 (Frontier Savings Bank) — transcribed without a date

Form 1098 #1 (Prairie Home Mortgage Co.) — transcribed without a date

Form 1098-E #1 (Federal Direct Loan Servicing) — transcribed without a date

("Transcribed without a date" is expected — I did not populate transcribed_on in the TOML; a real filer typing the same data by hand in the TUI would have it prompted.)

2.7 income project — the pre-export oracle check

```

$ btctax --vault v.pgp income project --year 2024
{
  "tax_year": 2024,
  "row": {
    "filing_status": "Married/Joint", "w2_income": 800000.0,
    "taxable_interest": 2500.0,
    "long_term_capital_gains": 55653.52, "state_income_tax": 40000.0,
    "real_estate_tax": 12000.0,
    "mortgage_interest": 40000.0, "standard_or_itemized": "Itemized",
    "hsa_deduction": 2000.0,

```

```

    "dependents": [
      {"age": 4, "credit": "child_tax_credit",
"eic_qualifying_child": true},
      {"age": 8, "credit": "child_tax_credit",
"eic_qualifying_child": true},
      {"age": 12, "credit": "child_tax_credit",
"eic_qualifying_child": true},
      {"age": 18, "credit":
"credit_for_other_dependents","eic_qualifying_child": true}
    ],
    "age_head": 39, "age_spouse": 37, "blind_head": false,
"blind_spouse": false
  },
  "refused": null,
  "not_carried": [
    {"leaf": "form_1098[0].box2_outstanding_principal", "amount":
"680000", "because": "NotInTheOracleRow", ...},
    {"leaf": "form_1098e[0].box1_interest", "amount": "2700",
"because": "NotInTheOracleRow",
      "note": "$221 student loan interest ... both engines model it and
the oracle row has no field, so
      a filer with a 1098-E is projected without the deduction"},
    {"leaf": "w2s[0].box12[0].amount", "amount": "3000", "because":
"NotInTheOracleRow"},
    {"leaf": "w2s[0].box2_fed_withheld", "amount": "200000",
"because": "PaymentOrWithholding"},
    {"leaf": "w2s[0].box3_ss_wages", "amount": "168600", "because":
"NotInTheOracleRow"},
    {"leaf": "w2s[0].box4_ss_withheld", "amount": "10453.20",
"because": "PaymentOrWithholding"},
    {"leaf": "w2s[0].box5_medicare_wages", "amount": "800000",
"because": "NotInTheOracleRow"},
    {"leaf": "w2s[0].box6_medicare_withheld", "amount": "17000",
"because": "PaymentOrWithholding"}
  ],
  "not_carried_from_the_ledger": []
}

```

"refused": null — the household is fully representable to both oracle engines (nothing in it hits a projection refusal); the not_carried list is the tool's own honest census of what the oracle row format cannot express (withholding lines, box-12 codes, the acquisition-debt ceiling), each with a one-line reason. This is expected/correct — informational, not a finding.

2.8 export-irs-pdf — the wall (Finding #2, the worst of the day)

```
$ btctax --vault v.pgp export-irs-pdf --out irs --tax-year 2024
error: IRS form fill: geometric read-back FAILED (mis-mapped cell):
form1[0].Page2[0].f2_04[0]: x-center 446.0 not in column 1 cluster
(504.0, 576.0) (mis-mapped column)
```

Exit code 2. **No irs/ directory was created — zero bytes written, no packet at all.** Reproduced twice, deterministically. f2_04 is mapped in crates/btctax-forms/forms/2024/f1040.map.toml:75 as line18 = "topmostSubform[0].Page2[0].f2_04[0]" — Form 1040 line 18 (tax after Schedule 2 Part I), an entirely ordinary money line present on any full return with tax due; nothing about my household's data is exotic (no AMT, no unusual credits). The tool's own export-irs-pdf --help explains the mechanism: *"Every written value is read back GEOMETRICALLY against the blank PDF's own field coordinates and the fill FAILS CLOSED on any mis-placement — a wrong tax form is never written."* — a deliberate safety property (never silently emit a misfiled PDF), but for this household it means the safety net itself is what stops the return: the mapped field's real on-page x-position (446.0) does not fall inside the money-column band the code expects (504.0–576.0) for that row, so the read-back check that is supposed to catch a wrong VALUE instead caught what looks like a wrong CELL COORDINATE in the line-18 map entry itself. No override flag exists (--forms is documented as *"ignored on a full-return year"*).

2.9 The car-loan probe (Finding #1) — an isolated scratch vault, walkA/ .. /probe-carloan/

The task's own household includes "a car loan with interest," and the walk-A year note says TY2024 has no such line. I tried it anyway, to see what the tool says. Copied household.toml, appended:

```
[[schedule_1a.vehicles]]
description = "2025 pickup truck, financed"
interest_paid = "1850"
loan_originated_after_2024 = true
loan_originated_by_you = true
proceeds_used_to_purchase = true
personal_use = true
secured_by_first_lien = true
original_use_starts_with_you = true
is_applicable_vehicle_class_under_14000_lbs = true
final_assembly_in_us = true
excludes_negative_equity = true
```

```
$ btctax --vault p.pgp income import --year 2024 --file
household_with_carloan.toml
Imported full-return inputs for tax year 2024.
```

No refusal, no warning, exit 0. income show confirms the row is stored verbatim:

```
$ btctax --vault p.pgp income show --year 2024 | jq .schedule_1a
{
  "tips": null, "overtime": null,
  "vehicles": [ { "description": "2025 pickup truck, financed",
"interest_paid": "1850", ... } ]
}
```

Reading the source (not modifying it): `Schedule1A::compute` (crates/btctax-core/src/tax/return_1040.rs:2435) returns `None` for any year outside 2025–2028 — its own comment says “*a TY2024 return yields zero BECAUSE THE FORM HAS NO LINE, not because a literal says so.*” The \$1,850 is accepted at import, carried in storage, and **silently discarded at computation with no advisory anywhere** (grepped `advisories.rs` for “vehicle”/“car loan”/“Schedule 1-A” — none exists for this case). Contrast every *other* out-of-scope item in this tool (1099-R, a rental, a K-1, an ISO exercise, an over-limit mortgage...), each of which gets an explicit refusal naming the exit, or a printed advisory sizing the gap. This one gets neither.

3. Walk B — TY2025, short probe

```
$ btctax --vault v.pgp init --key-backup key-backup.asc
Vault passphrase: *****
Confirm passphrase: *****
Initialized vault v.pgp (key backed up to key-backup.asc)
```

```
$ btctax --vault v.pgp income import --year 2025 --file
household_2025.toml
note: TY2025 – preparing (18 forms; TaxTable yes; full-return params
no; 1099-DA proceeds) – these
inputs are stored now; `report --tax-year 2025` computes the crypto
delta on a stored `tax-profile`
and reports the full return as NOT COMPUTABLE (keeping the inputs)
until full-return parameters for
2025 are bundled.
Imported full-return inputs for tax year 2025.
```

Import **succeeds** — this is not the wall. `income answer --year 2025` also works, in `author-only` mode:

```
$ btctax --vault v.pgp income answer --year 2025
Answering full-return questions for tax year 2025:
interview: 41 question(s) still to answer · return: NOT computable –
authoring and saving work;
computing and committing wait for the TY2025 package – TY2025 –
preparing (18 forms; TaxTable yes;
full-return params no; 1099-DA proceeds)
```

— Step 0: your ledger, for 2025 —

nothing outstanding: no blockers, no imports, no venue without an answer.

— The answer panel (before) —

BLOCKING (41) – commit waits on these: ...

income show --year 2025 confirms the \$800,000 wages and every other figure are genuinely stored.

3.1 The wall — report --tax-year 2025

```
$ btctax --vault v.pgp report --tax-year 2025
error: usage: tax year 2025 has full-return inputs, but full-return
computation is not available for
it in this build – TY2025 – preparing (18 forms; TaxTable yes; full-
return params no; 1099-DA
proceeds). The inputs are KEPT and will compute when the year's
package is bundled. To fall back to a
raw `tax-profile` for 2025 instead, run `income clear --year 2025`
(this DISCARDS the stored inputs,
including any computed carryover on them).
```

Exit code 2. This is the expected, documented stop: the message names exactly what happened (no TY2025 tax-table/full-return package is bundled in this build), what is preserved (the inputs — they are not lost), and the one available way forward (income clear, explicitly framed as destructive). **Stopped here, as instructed** — this is the wall the task described in advance, and it behaves exactly as documented.

3.2 One step further anyway — export-irs-pdf --tax-year 2025 (Finding #5)

Curious whether the crypto side alone would export, I tried the packet command too (still short of “fighting” the wall — this is a different command, not a workaround for the one above):

```
$ btctax --vault v.pgp export-irs-pdf --out irs --tax-year 2025
Filled IRS forms for tax year 2025 →
  irs/f8949.pdf
  irs/schedule_d.pdf
```

▲ NOT AUTHORISED FOR FILING. btctax is a mechanical calculator. ...
note: Schedule D line 17 was answered from the printed lines 15 and 16. Lines 18-22 are OUT OF SCOPE
and left blank ...

Exit 0, “success,” two files — with **zero mention** that full-return inputs (wages, mortgage, HSA, 4 dependents) were ever imported for this year, or that none of it is in the output. Reading the dispatch (crates/btctax-cli/src/cmd/admin.rs:887-917, not modified): the “answers-driven crypto slice” branch’s condition is

`!ri.broker_reporting.0.is_empty()` — the TY2026+ Form 1099-DA broker-reporting table specifically, which my household never touches (broker reporting doesn't exist before TY2026) — so this run took the plain, oldest crypto-only path, identical byte-for-byte to what `export-irs-pdf` would have produced if `income import` had **never been run** for 2025 at all.

4. FINDINGS

Ranked by how much each would matter to a real filer at a deadline.

1. [CRITICAL — blocks filing entirely] `export-irs-pdf` crashes for the whole TY2024 full return. What I wanted: the printed packet for a fully-answered, fully-computed, correct return (the interview completed with 0 blocking, 0 refusing). What happened, verbatim: `error: IRS form fill: geometric read-back FAILED (mis-mapped cell): form1[0].Page2[0].f2_04[0]: x-center 446.0 not in column 1 cluster (504.0, 576.0) (mis-mapped column), exit 2, no output directory, no bytes written`. Class: a refusal (fail-closed by design) that behaves like a crash from the filer's chair — a cryptic internal coordinate message, no guidance, no workaround (— forms is documented as ignored on this path). What I dropped: nothing — there was nothing to drop; the return itself is complete and correct (`report/income` project both show it), only the PDF-writing step fails. This is a mainstream household shape (MFJ, wages, mortgage, 4 kids, some Bitcoin gains) — not an edge case I constructed to break something.

2. [Important — silent, no signal at all] TY2024 has no field for car-loan interest, and says nothing about it. What I wanted: enter \$1,850 of 2024 car-loan interest. What happened: `income import` accepted a fully-detailed `[[schedule_1a.vehicles]]` row with **exit 0 and no message of any kind**; the figure is stored (`income show` echoes it back) but the computation discards it entirely (`Schedule1A::compute` returns `None` before 2025, “because the form has no line”), and grepping the advisories module turns up nothing that ever mentions it. Every *other* out-of-scope item in this codebase gets an explicit refusal or a printed advisory sizing the gap; this is the one silent exception I found. What I dropped: the car loan, per the task's own year note — but a filer who didn't already know that would have no way to learn it from the tool.

3. [Important — misleading “success” near a deadline] `export-irs-pdf --tax-year 2025` silently produces a 2-file crypto-only packet, with no indication the household's full-return data (already imported) isn't in it. What happened: Filled IRS forms for tax year 2025 → `irs/f8949.pdf` `irs/schedule_d.pdf`, exit 0, no warning. The dispatch's “answers-driven” branch checks only the TY2026+ broker-reporting table, so a TY2025 filer who carefully entered wages/dependents/mortgage/HSA gets the exact same 2-form output as someone who never ran `income import` at all, with nothing in the CLI output distinguishing the two cases.

4. [Minor — UX friction, not incorrect] Dependent §152 gates are re-asked in full, every single `income answer` session, even when every leaf was already truthfully answered via the imported TOML. For 4 dependents that is ~64 additional y/n/date prompts per session (bare Enter re-confirms the value shown as “currently ...”). Correct, but tedious, and nothing in the prompt explains *why* an already-answered gate is being asked again.

5. [Minor — confusing message] The “before” answer panel undercounts. The panel printed “BLOCKING (36)” at the start of the session, but two more questions (InterestOrDividendsWithout1099, StateRefundWithout1099g) became live and were asked mid-session, triggered by my own earlier census answers within the *same* pass. By design (documented in-source as the M-3 seam-review fix), but the panel a filer reads to gauge “how much is left” is a moving target they aren’t warned about.

6. [Minor — real but narrow ambiguity] Form 5498-SA’s census row cannot honestly represent “haven’t received it yet, and don’t need it.” Form 5498-SA is legally issued only after the April filing deadline (contributions can be made up to the return’s due date), so an early filer can never truthfully answer “Yes” to the census row, yet the true state is “not yet received,” not “will never receive.” The census is a plain y/n at the keyboard (unlike the tri-state the spec itself describes for “not yet received”), so I answered “n” — a defensible but not literally true testimony given the household doesn’t need the form to compute its own HSA figures.

Observed and correct (not findings, per the task’s own list): student loan interest deducting \$0 despite a real \$2,700 paid (\$221 phases out completely at this income); Child Tax Credit / Credit for Other Dependents both \$0 despite 4 dependents, one under 6 and one 17+ (§24(b) phases the CTC out entirely at \$800k MFJ) — and the row-(7) grid still correctly differentiates Jasper (18, “Credit for other dependents”) from the other three (under 17, “Child tax credit”), proving the age-cutoff logic works even though the dollar amount is moot here.

5. DID IT FILE?

Walk A (TY2024): NO. The interview completed fully (115 questions answered, 0 blocking, 0 refusing), report `--tax-year 2024` shows a complete, internally-consistent, correct return (AGI \$856,154, itemized deduction \$50,000 — \$40,000 mortgage interest + the \$10,000-capped SALT — taxable income \$806,154, tax \$214,941, +\$4,950 Additional Medicare, +\$2,210 NIIT, \$0 AMT — verified by hand: AMTI \$816,153.52 = taxable income + the \$10,000 SALT add-back; exemption \$133,300 is the correct 2024 MFJ figure; TMT \$182,094.70 < regular tax so AMT is correctly \$0 — total tax \$222,101, withholding \$205,400 = \$200,000 federal + \$5,400 imputed Additional-Medicare withholding, amount owed \$16,701), and income project shows the household is fully representable to both oracle engines (“refused”: null). **But export-irs-pdf crashes** (Finding #1) — no PDF, no packet, no manifest, zero bytes written. The computation is done and correct; nothing printable came out of it.

Walk B (TY2025): NO, and expected not to. report `--tax-year 2025` refuses cleanly, by design, naming exactly why (no TY2025 tax-table/full-return package bundled) and what to do (wait, or income clear). `export-irs-pdf --tax-year 2025` does produce two files (f8949.pdf, schedule_d.pdf), but they are the crypto-only slice and do not represent the household at all (Finding #3) — not a real partial filing.

6. WHAT SURPRISED YOU

- **init needs a real TTY and gives no hint why.** Piped stdin gets error: `io: No such device or address (os error 6)` with no suggestion to use a terminal or the BTCTAX_PASSPHRASE env var. Once I found that var in `main.rs`, everything

else scripted cleanly — but I had to go read the source to find it; nothing in `--help` mentions it. (Per this project’s own severity ruling, secret-handling friction never gates — noting it here as a surprise, not a finding.)

- **The interview’s actual question ORDER does not match the panel’s print order**, and isn’t the registry declaration order either — the document census came first in the live run, ahead of the dependent-status/foreign-account declarations the “before” panel lists first. Nothing wrong, just unexpected while scripting answers by keyword.
- **A census “No” can silently open a brand-new, unpreviewed question in the very same pass** — the “document-less income door” pattern (Finding #5) is a well-reasoned design (the form really does ask “did you have interest with no 1099”), but from the driver’s seat it looks like the interview changed its mind about how many questions there are, mid-conversation.
- **The dependent grid asks you to reconfirm children you already fully described**, every time, forever (as far as I could tell in one session) — I expected “already answered” fields to be skip- or summary-only.
- **The PDF export’s fail-closed geometry check is a genuinely good idea** (never silently mis-file a form) — and it is also, in this run, the single thing standing between a complete correct return and a printed packet, for an entirely ordinary household. The safety net and the blocker are the same code path.
- **export-irs-pdf never told me it was falling back** to a 2-file crypto slice on TY2025 even though I had just imported a whole household’s worth of full-return data for that exact year in the same session. I had to read the dispatch source to learn the crypto-slice path was chosen because of an unrelated TY2026-only field being empty, not because of anything about TY2025 readiness per se.